Introduction to Macroeconomics (S102016CR)

Spring Semester

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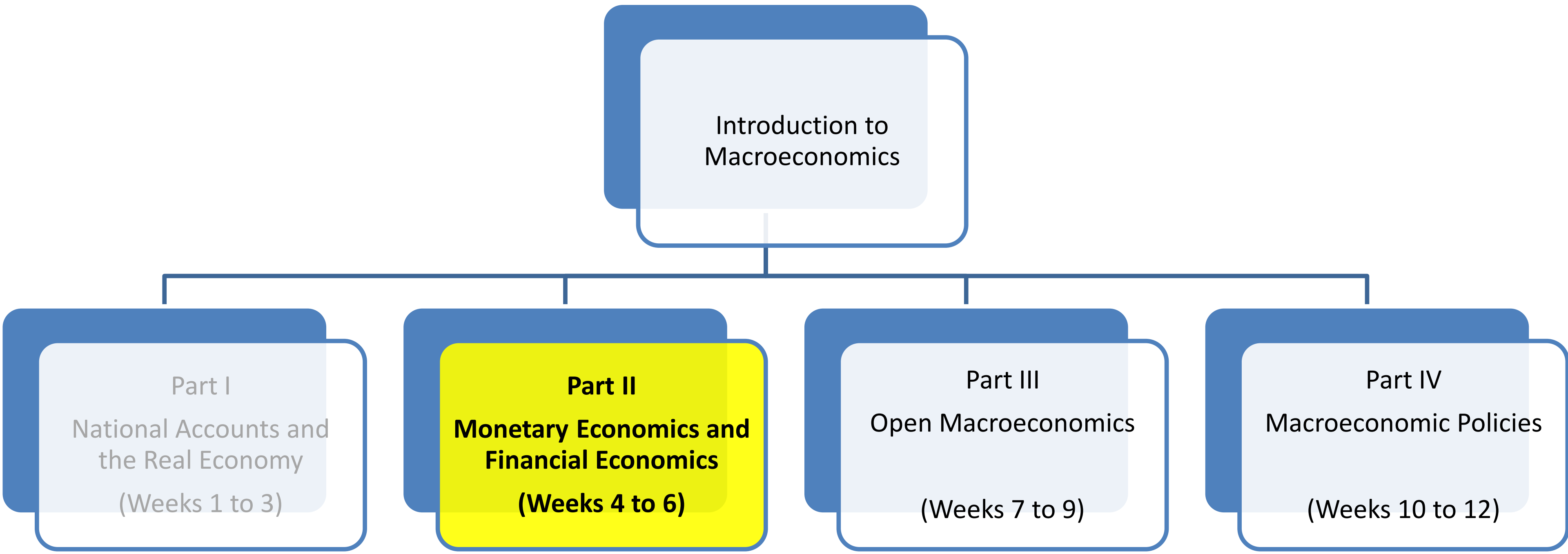
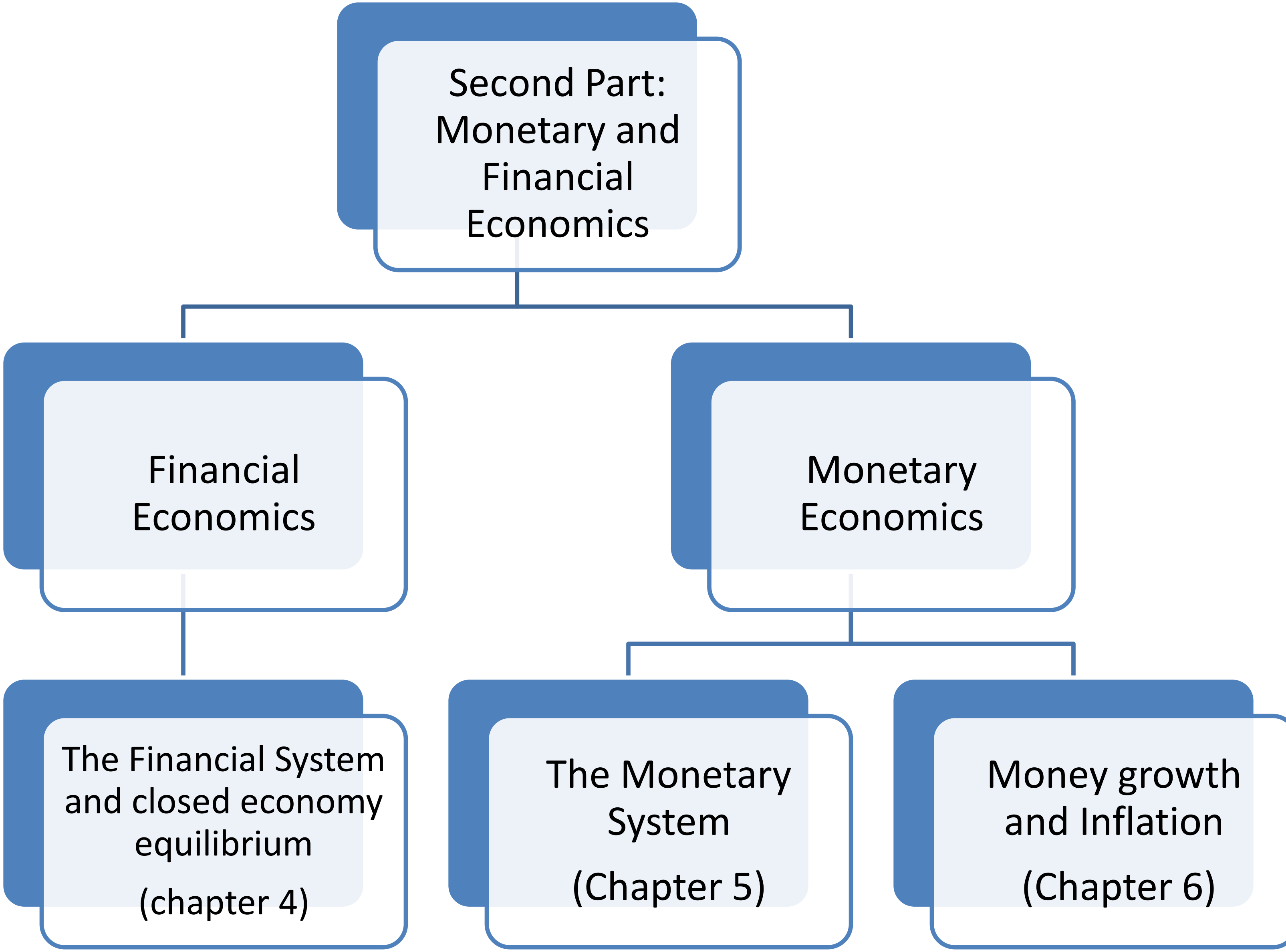
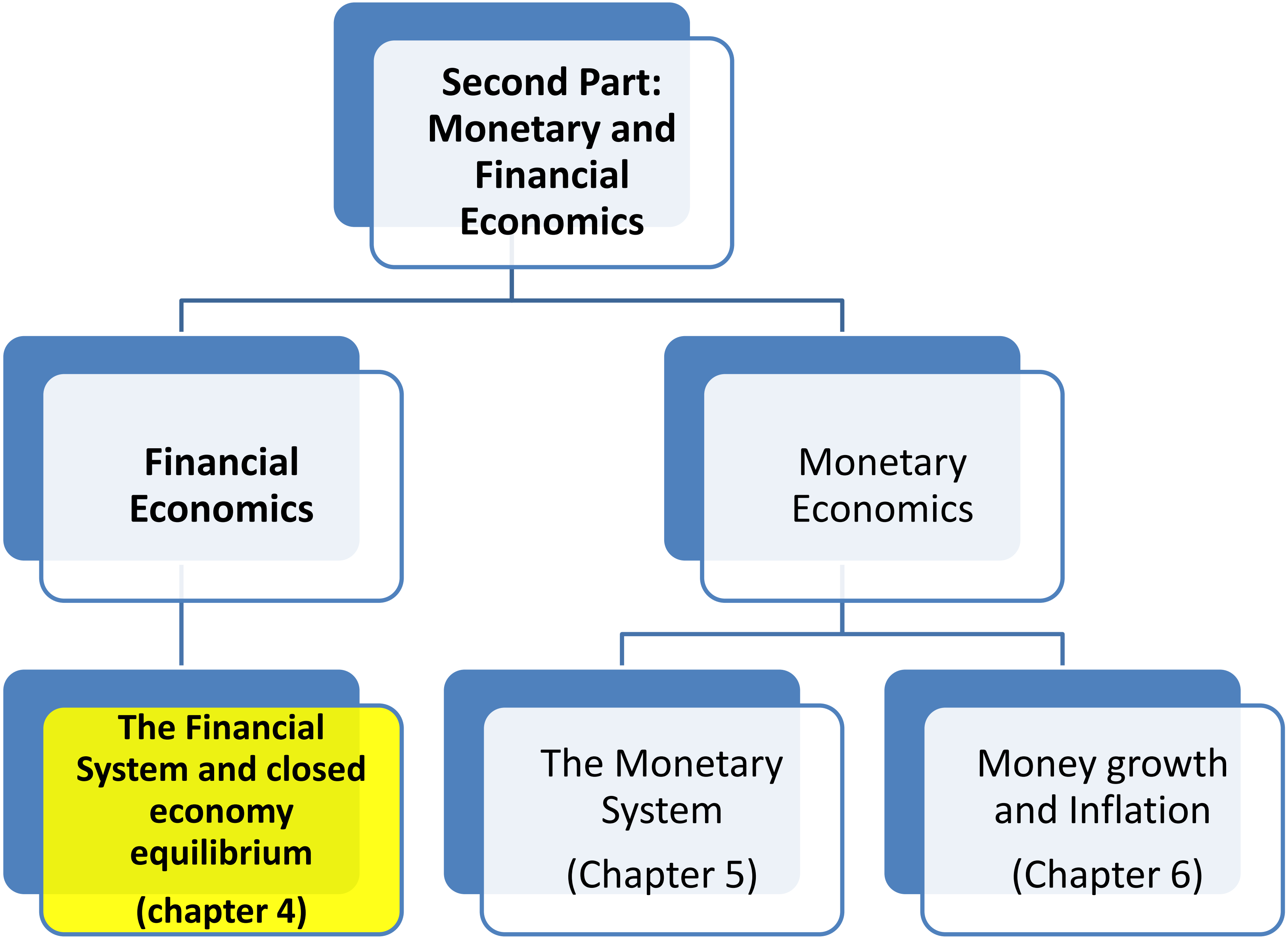

Introduction to Macroeconomics
Part I
National Accounts and the Real Economy
(Weeks 1 to 3)
Part II
Monetary Economics and Financial Economics
(Weeks 4 to 6)
Part III
Open Macroeconomics
(Weeks 7 to 9)
Part IV
Macroeconomic Policies
(Weeks 10 to 12)



Second Part: Monetary and Financial Economics
Financial Economics
Monetary Economics
The Financial System and closed economy equilibrium
(chapter 4)
The Monetary System
(Chapter 5)
Money growth and Inflation
(Chapter 6)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Week 4:
The Financial System and closed economy equilibrium

- The financial system consists of a group of institutions in the economy that help match one person's savings with another person's investments.

- It moves the economy's scarce resources from savers to borrowers.

- The objective of this chapter is to:
  - describe the functioning of the financial system
  - the relationship between aggregate savings and investment, which determines the closed economy macroeconomic equilibrium
  - the role of the government in ensuring the well-functioning of the financial system to promote long-term growth

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a) Financial Institutions

b) Aggregate Savings and Investment and closed economy equilibrium

c) What role for the Government?

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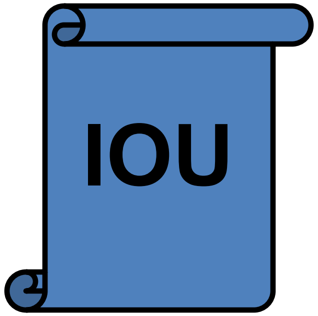
a. Financial Institutions

- Financial institutions can be grouped into two different categories: financial markets and financial intermediaries.

- *Financial markets* are the institutions through which savers can directly provide funds to borrowers.
  - Bond Market
  - Stock Market
  - Stocks and bonds are called securities because they represent secure or “asset-based” claims by investors against issuers. Bondholders have preferential treatment when a company cannot repay.
  - Primary versus secondary financial markets: Is the seller the issuer?

- *Financial intermediaries* are financial institutions through which savers can indirectly provide funds to borrowers.
  - Banks
  - Mutual Funds

- The Bond Market
  - A *bond* is a certificate of indebtedness that specifies obligations of the borrower to the bondholder of the bond. They are issued by issue by the government, provinces, and firms.
  - Allows institutions to debt finance their projects
  - Characteristics of a Bond
    - *Term*: The length of time until the bond matures.
    - *Credit Risk*: The probability that the borrower will fail to pay some of the interest or principal.



- These two characteristics will determine the interest rate spread that the borrower will pay the bondholder above the risk-free interest rate:
  - « junk bonds » will have to pay a higher interest rate than Swiss government bonds
  - 10-year bonds will have to pay a higher interest rate than three months bonds

- The price of a bond will then depend on the credit risk
  - If the credit risk increases, the price of the bond declines, as you will receive the same nominal payments as before but taking a higher degree of risk. The bond’s price will be below its face value to compensate for this.

    Yield (interest rate including credit risk) = Payments / Bond price

    → Bond price = Payments / Yield

    The bond price will decline if the yield increases because of higher risk.

    The bond's price is not necessarily equal to its face value (or par value); therefore, the yield varies with the bond's price.

- The Stock Market

  Stock represents a claim to partial ownership in a firm and is therefore, a claim to the profits that the firm makes. Raising funds through the sell of stocks is called equity financing.

- The payments to stockholders are called dividend and they correspond to their share of ownership * profits paid.

- Compared to bonds, stocks offer both higher risk and potentially higher returns.
  - The average yield of stocks in the S&P stock market between 1927 and 2027 is around 10% (7% real terms).
  - The average 10-year US Treasury bond yield was 5% during the same period

- Stocks are traded on exchanges such as the Swiss Stock Market, London Stock Exchange, Nasdaq, New York Stock Market, etc...

- *Financial intermediaries* are financial institutions through which savers can indirectly provide funds to borrowers.
- Banks
  - take deposits from people who want to save and use the deposits to make loans to people who want to borrow.
  - pay depositors interest on their deposits and charge borrowers slightly higher interest on their loans.
- Investment Funds
  - An *investment or mutual fund* is an institution that sells shares to the public and uses the proceeds to buy a portfolio of various types of stocks, bonds, or both.
    - They allow people with small amounts of money to diversify across industries, countries, etc…combined with professional management
    - The price of a mutual fund share or as it is called the Net Asset Value = current market value of the fund’s investment /# shares

- Other Financial Institutions
  - Credit unions which are cooperatives owned and run by its clients ("caisse populaire" or "banque populaire" in french).
  - Pension funds which invest the savings of workers in order to ensure an income at the moment they retire
  - Insurance companies which invest the fees paid by clients to pay the client in case of an insured event or at retirement
  - Pawnbrokers where borrowers with bad risk ratings can borrow money and leave some valuable item such as a watch as security
  - Loan sharks very high interest rates unsecured loans often backed by threats or blackmail

"Will you be passing on the cut in the Bank of England base rate?"

Pricing an asset

- The price of an asset depends on the willingness to pay for that asset
- And the willingness to pay depends on the expectation of future profits
- The determinants of future profits are what we call the “fundamentals” driving future profits and, therefore, the asset’s price.

If financial markets are efficient:

- All public information available regarding the fundamentals is incorporated in the price of the asset
- If there is a difference between the value indicated by the analysis of fundamentals and the price observed in the market, then there is a price opportunity that will be taken if markets are efficient
- This leads to the observation that asset prices follow a random walk that follows "news" brought into the market (sometimes "good" and sometimes "bad")

But irrational behavior, "animal spirits" sometimes lead to inefficient markets

- Tech bubble at the end of the 1990s
- US House bubble in the 2000s
- NFTs and crypto bubble in 2022-2025

When investors believed that prices would keep on rising forever instead of basing their pricing decisions on fundamentals

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b. Aggregate Savings and Investment and closed economy eq.

Recall that GDP is both total income in an economy and total expenditure on the economy's output of goods and services:

$$Y = C + I + G + X\text{-}M$$

Assume a closed economy – one that does not engage in international trade:

$$Y = C + I + G$$

Now, subtract C and G from both sides of the equation:

$$Y – C – G = I$$

The left side of the equation is the total income in the economy after paying for consumption and government purchases and is called *national saving,* or just *saving (S):*

$$S = I$$

- We can decompose national savings as follows:

$$S = Y - C - G$$

$$S = \underbrace{(Y - T - C)}_{\text{Private savings}} \quad + \quad \underbrace{(T - G)}_{\text{Public Savings}}$$

Savings = Private savings + Public Savings

$$S = S^P + (T - G)$$

- Thus, *S=I* becomes

$$S^P + (T - G) = I$$

  - Investment can be funded through private savings or government surplus.
  - Or as we will see later, a government deficit will reduce the amount of private savings available to invest. A government deficit crowds out private investment

- Rearrange the previous expression to obtain

$$S^P - I = G - T$$

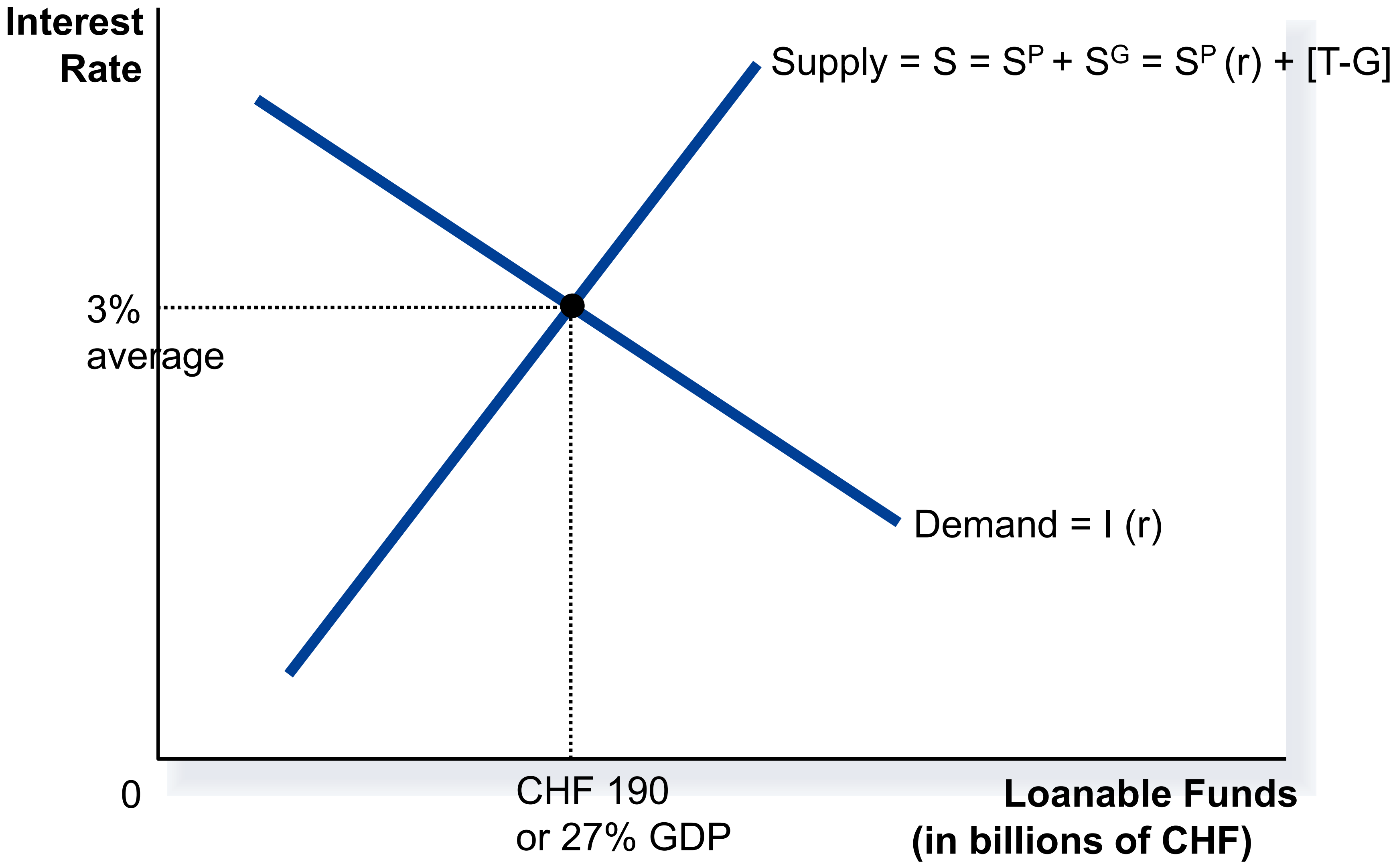
- Government's deficit (G-T>0) is directly linked with private savings and investment.

- Excess savings with respect to investment ($S^P > I$) lead to a government deficit (G>T) which is funded by excess savings

- Excess investment with respect to savings ($S^P < I$) *lead to a government surplus* (G<T), which helps fund the excess investment

- Financial markets coordinate the economy's savings and investment in the market for loanable funds.
- The *market for loanable funds* is the market in which those who want to save supply funds and those who want to borrow to invest demand funds.
- Loanable funds refer to all income people choose to save and lend out, rather than use for consumption.
- The supply of loanable funds comes from people with extra income they want to save and lend out (or government surplus).
- The demand for loanable funds comes from households and firms that wish to borrow to make investments.
- The interest rate is the price of the loan.
- It represents the amount borrowers pay for loans and lenders receive on their savings.
- The interest rate in the market for loanable funds is the real interest rate.
- The equilibrium of the supply and demand for loanable funds determines the *real interest rate*.

The market for loanable funds and closed economy equilibrium



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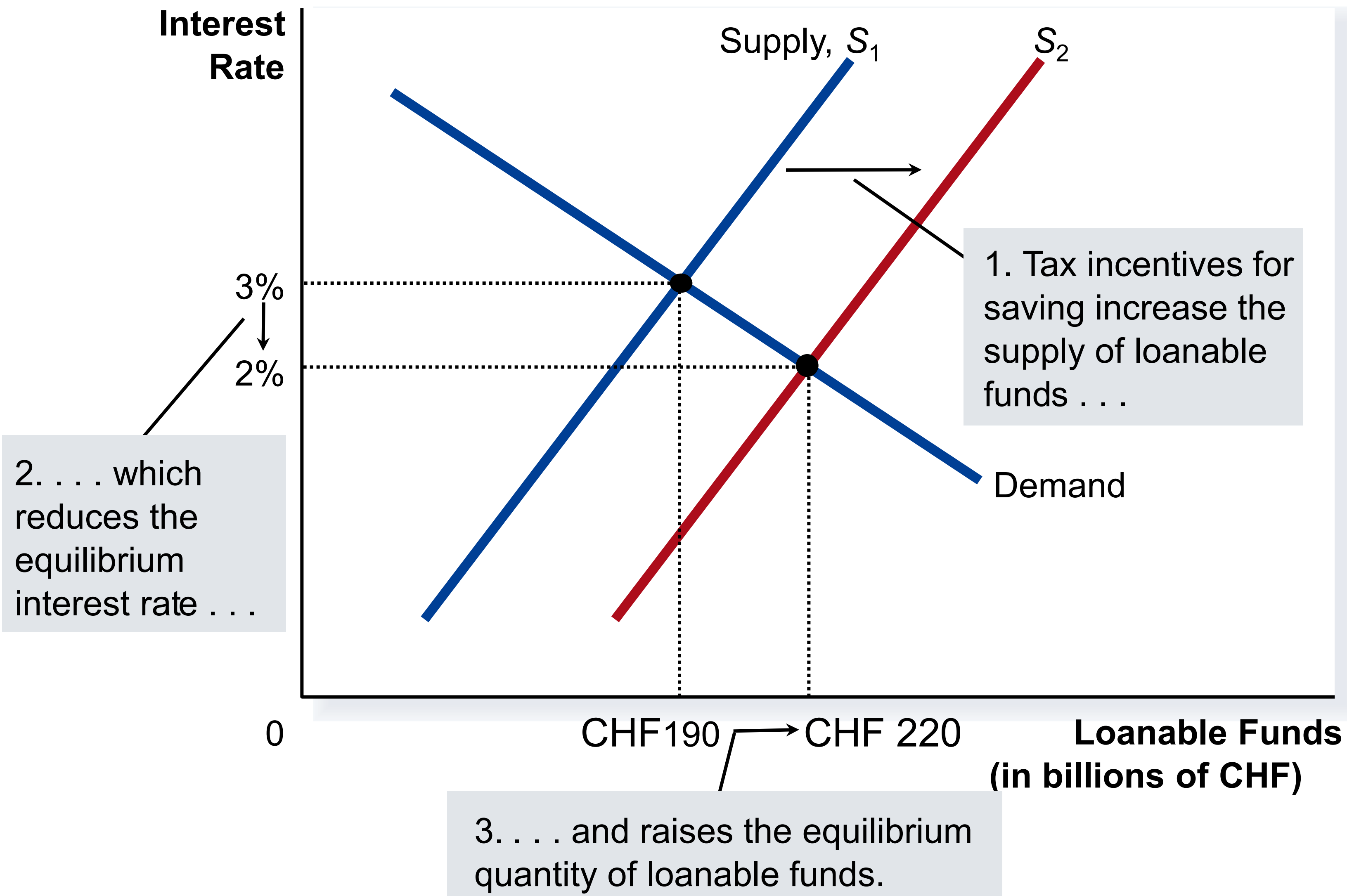
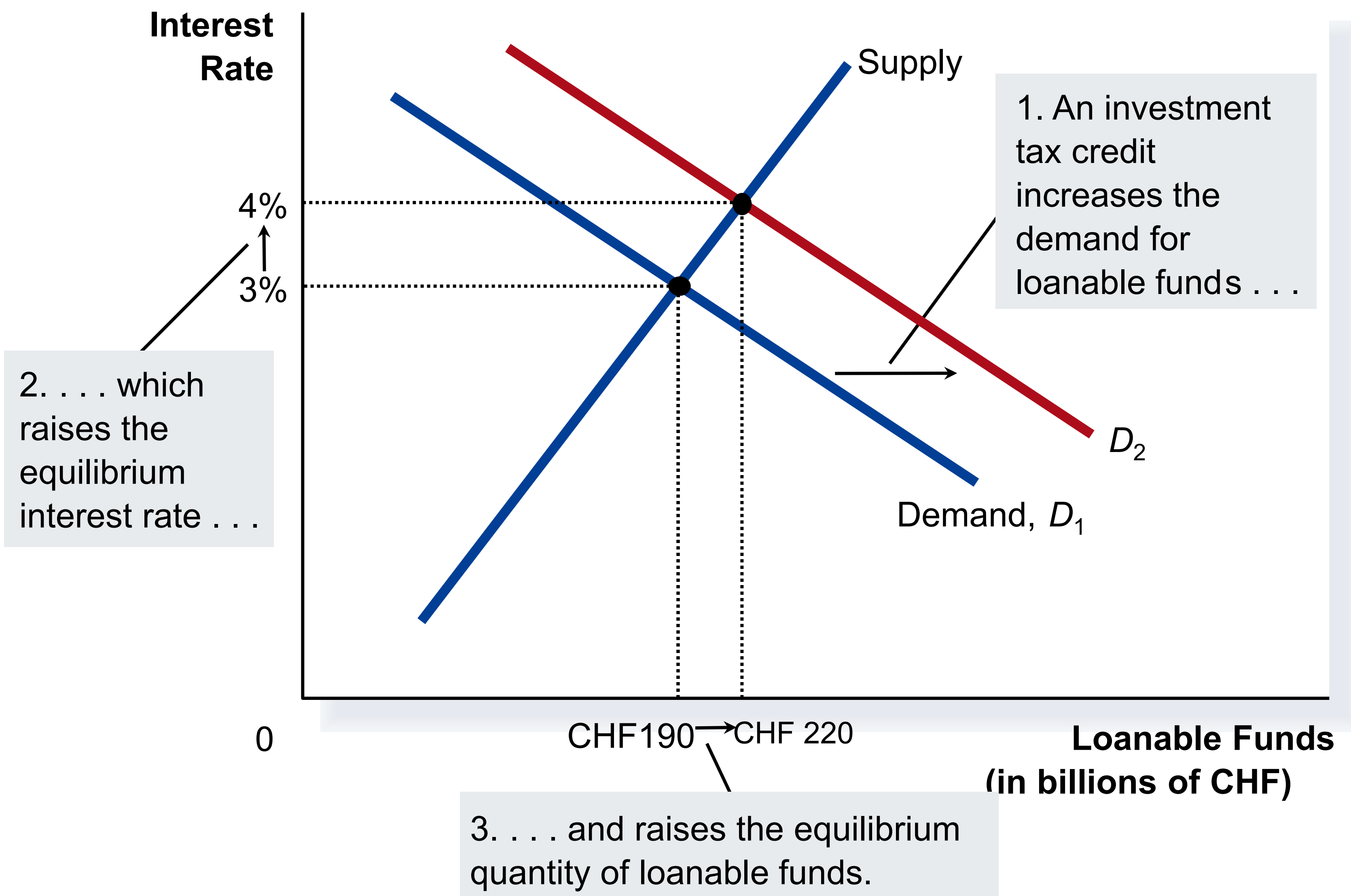
c. What role for the government?

- A determinant of productivity and long-term growth is the amount of capital available in the economy. One of the government's objectives is then to promote capital formation (investment)
- Government Policies that affect savings and investment
  - Taxes on interest income and savings: Taxes on interest income reduce the future payoff from current savings and, as a result, reduce the incentive to save. A tax decrease on interest income increases the incentive for households to save at any given interest rate. The supply of loanable funds increases
  - Taxes on income and investment: An investment tax credit increases the incentive to borrow. The demand for loanable funds increases
  - Government budget deficits: Government borrowing to finance its budget deficit reduces the supply of loanable funds available to finance investment by households and firms. This fall in investment is referred to as *crowding out*.
    - The deficit borrowing crowds out private borrowers trying to finance investments.

Tax incentives for savings



Tax incentives for investment



But why subsidize savings or investments?

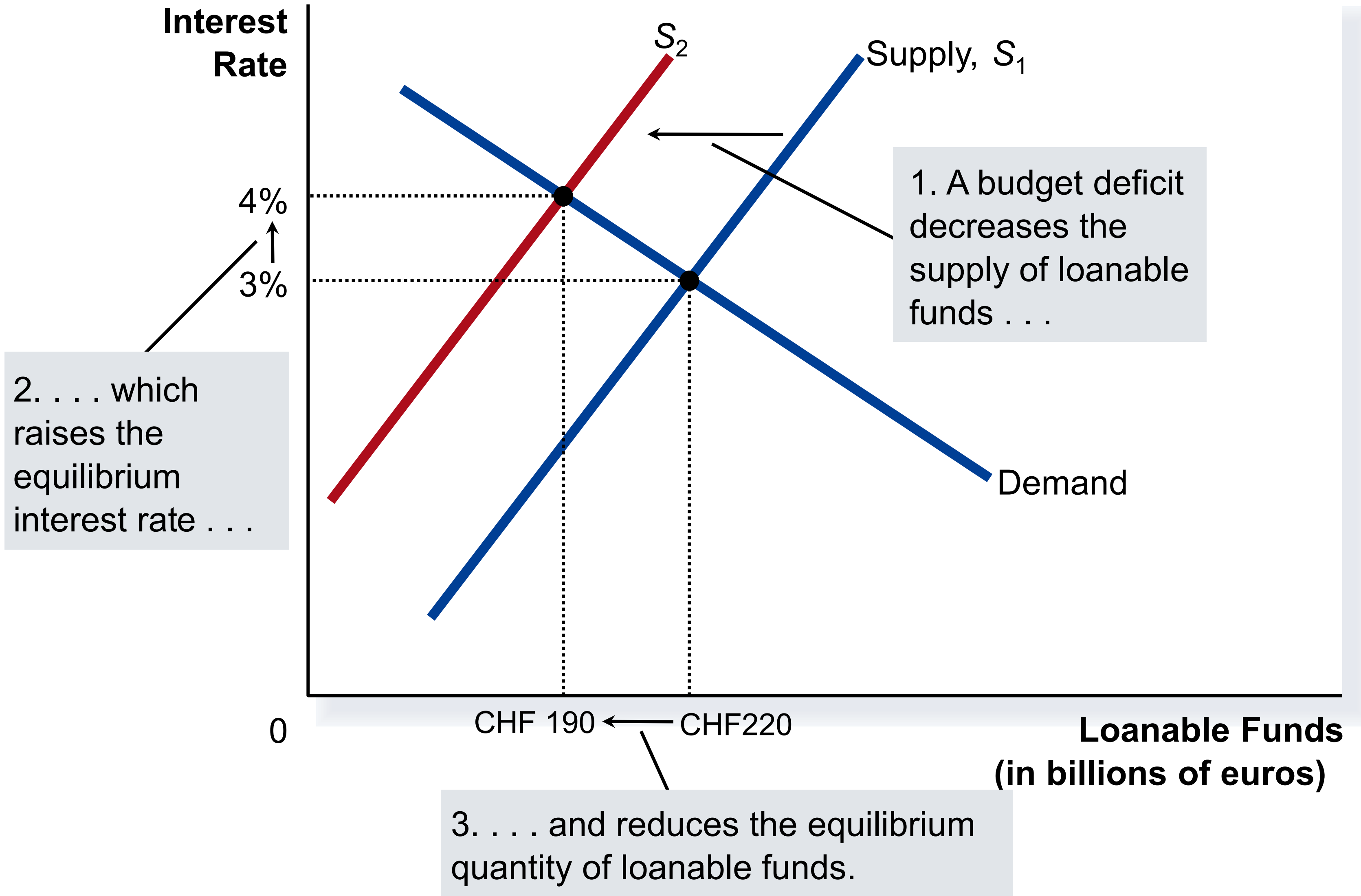
- To boost capital accumulation, but there are a few issues

- One problem with subsidising savings or investment is that government income will have to come from somewhere else (tax on consumption or labour income), which is regressive, as wealthy households tend to consume a smaller share of their income than poor households or have their share of labour income is relatively smaller than the one of poor households. Investments are largely undertaken by wealthy households.

- Another problem with tax incentives for savings is that savings may not be very sensitive to interest rates, as the income and substitution effect work in the opposite direction: a higher interest rate implies a higher income for a given level of savings and, therefore, may push individuals to save less, where the substitution effect calls for more saving and less consumption. Empirically, the two effects seem to cancel each other.

- Similarly, investment may not be very interest rate elastic during an economic crisis (so not recommended during downturns)

Budget deficits



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- The financial system is made up of financial institutions such as the bond market, the stock market, banks, and investment funds.
- All these institutions act to direct the resources of households who want to save some of their income into the hands of households and firms who want to borrow.
- National income accounting identities reveal some meaningful relationships among macroeconomic variables.
- In a closed economy, national saving must equal investment. It determines the closed economy equilibrium
- Financial institutions attempt to match one person's saving with another person's investment.
- The supply and demand for loanable funds determine the interest rate.
- The supply of loanable funds comes from households who want to save some of their income.
- The demand for loanable funds comes from households and firms who want to borrow for investment.

- National saving equals private saving plus public saving.
- A government budget deficit represents negative public saving and, therefore, reduces national savings and the supply of loanable funds.
- When a government budget deficit crowds out investment, it reduces the growth of productivity and GDP.